

ISE 503 Course Project

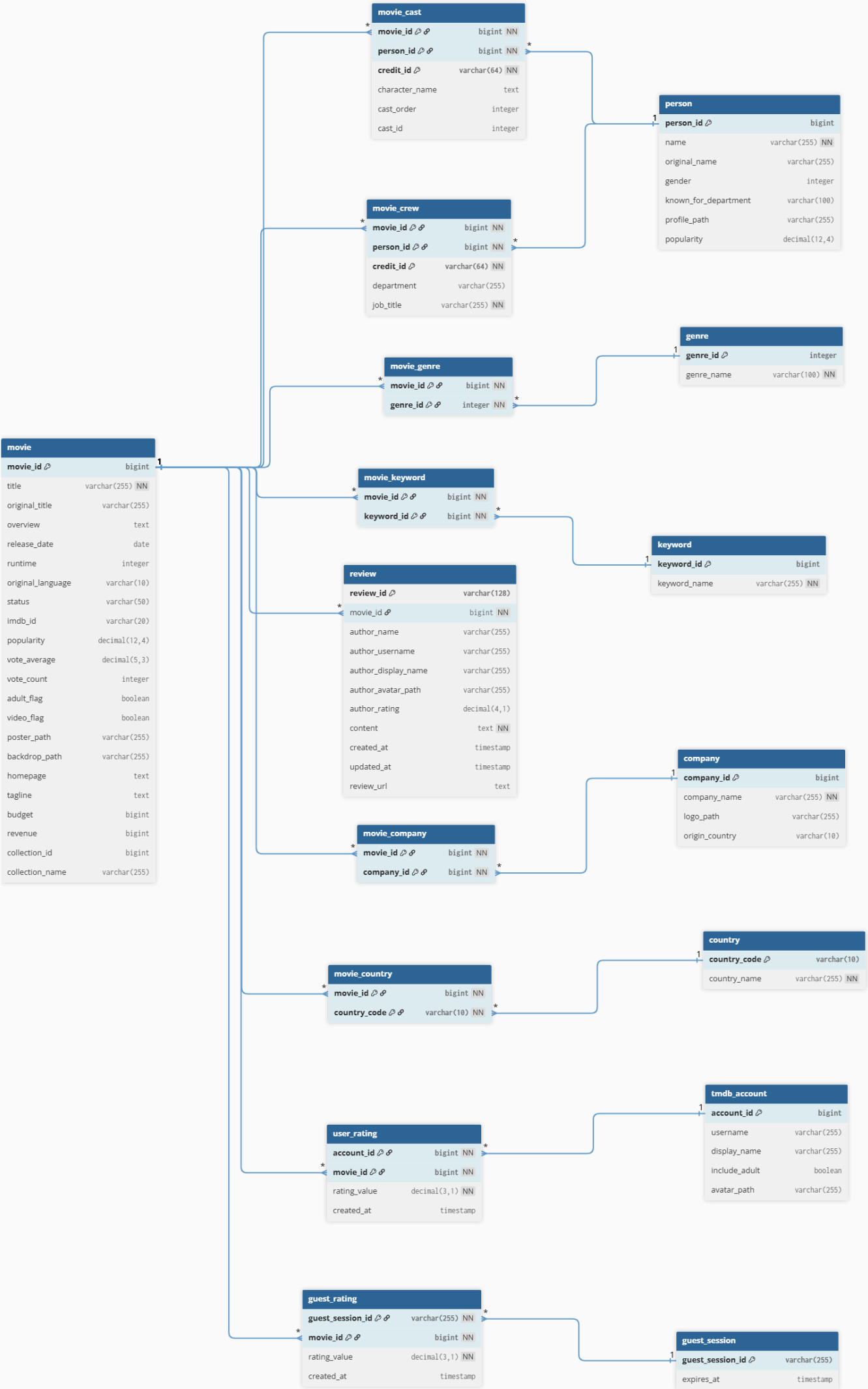
Group: 10

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The schema diagram on the next page was created with DBML (Database Markup Language) on <https://dbdiagram.io/>

Reference: <https://docs.dbdiagram.io/>

The PostgreSQL DDL script used to define and create the database tables is also attached.



db\init\01_create_tables.sql

```

1  -- =====
2  -- TMDb Movie Recommender - PostgreSQL Schema
3  -- =====
4
5  -- -----
6  -- Core entity: MOVIE
7  -- -----
8  CREATE TABLE IF NOT EXISTS movie (
9      movie_id          BIGINT          PRIMARY KEY,
10     title              VARCHAR(255)    NOT NULL,
11     original_title     VARCHAR(255),
12     overview           TEXT,
13     release_date       DATE,
14     runtime            INT,
15     original_language  VARCHAR(10),
16     status             VARCHAR(50),
17     imdb_id            VARCHAR(20)     UNIQUE,          -- globally unique IMDB identifier
18     popularity         DECIMAL(12, 4),
19     vote_average       DECIMAL(5, 3),      -- TMDb returns values like 8.203
20     vote_count         INT,
21     adult_flag         BOOLEAN,
22     video_flag         BOOLEAN,
23     poster_path        VARCHAR(255),
24     backdrop_path      VARCHAR(255),
25     homepage          TEXT,
26     tagline            TEXT,
27     budget             BIGINT,
28     revenue            BIGINT,
29     collection_id      BIGINT,
30     collection_name    VARCHAR(255)
31 );
32
33 -- -----
34 -- GENRE + MOVIE_GENRE (M:N)
35 -- -----
36 CREATE TABLE IF NOT EXISTS genre (
37     genre_id          INT              PRIMARY KEY,
38     genre_name        VARCHAR(100)    NOT NULL UNIQUE    -- genre names are globally unique
39 );
40
41 CREATE TABLE IF NOT EXISTS movie_genre (
42     movie_id          BIGINT NOT NULL,
43     genre_id          INT     NOT NULL,
44     PRIMARY KEY (movie_id, genre_id),
45     FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE,
46     FOREIGN KEY (genre_id) REFERENCES genre(genre_id) ON DELETE CASCADE
47 );
48

```

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49  -- -----
50  -- PERSON + MOVIE_CAST + MOVIE_CREW
51  -- -----
52  CREATE TABLE IF NOT EXISTS person (
53      person_id          BIGINT          PRIMARY KEY,
54      name                VARCHAR(255)    NOT NULL,
55      original_name       VARCHAR(255),
56      -- 0=Unknown, 1=Female, 2=Male, 3=Non-binary/Other (TMDB occasionally uses 3)
57      gender              INT             CHECK (gender IN (0, 1, 2, 3) OR gender IS NULL),
58      known_for_department VARCHAR(100),
59      profile_path        VARCHAR(255),
60      popularity          DECIMAL(12, 4)
61  );
62
63  -- credit_id is part of the PK so an actor with multiple roles in one movie
64  -- (e.g. voice cast + on-screen) gets a separate row per credit
65  CREATE TABLE IF NOT EXISTS movie_cast (
66      movie_id           BIGINT          NOT NULL,
67      person_id          BIGINT          NOT NULL,
68      credit_id          VARCHAR(64)     NOT NULL,
69      character_name     TEXT,
70      cast_order         INT,
71      cast_id            INT,
72      PRIMARY KEY (movie_id, person_id, credit_id),
73      FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE,
74      FOREIGN KEY (person_id) REFERENCES person(person_id) ON DELETE CASCADE
75  );
76
77  CREATE TABLE IF NOT EXISTS movie_crew (
78      movie_id           BIGINT          NOT NULL,
79      person_id          BIGINT          NOT NULL,
80      credit_id          VARCHAR(64)     NOT NULL,
81      department         VARCHAR(255),
82      job_title          VARCHAR(255)    NOT NULL,
83      PRIMARY KEY (movie_id, person_id, credit_id),
84      FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE,
85      FOREIGN KEY (person_id) REFERENCES person(person_id) ON DELETE CASCADE
86  );
87
88  -- -----
89  -- KEYWORD + MOVIE_KEYWORD (M:N)
90  -- -----
91  CREATE TABLE IF NOT EXISTS keyword (
92      keyword_id         BIGINT          PRIMARY KEY,
93      keyword_name       VARCHAR(255)    NOT NULL UNIQUE -- keyword names are globally unique
94  );
95
96  CREATE TABLE IF NOT EXISTS movie_keyword (
97      movie_id           BIGINT          NOT NULL,
98      keyword_id         BIGINT          NOT NULL,

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99     PRIMARY KEY (movie_id, keyword_id),
100     FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE,
101     FOREIGN KEY (keyword_id) REFERENCES keyword(keyword_id) ON DELETE CASCADE
102 );
103
104 -- -----
105 -- COMPANY + MOVIE_COMPANY (M:N)
106 -- -----
107 CREATE TABLE IF NOT EXISTS company (
108     company_id      BIGINT      PRIMARY KEY,
109     company_name    VARCHAR(255) NOT NULL,
110     logo_path       VARCHAR(255),
111     origin_country  VARCHAR(10)
112 );
113
114 CREATE TABLE IF NOT EXISTS movie_company (
115     movie_id      BIGINT NOT NULL,
116     company_id    BIGINT NOT NULL,
117     PRIMARY KEY (movie_id, company_id),
118     FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE,
119     FOREIGN KEY (company_id) REFERENCES company(company_id) ON DELETE CASCADE
120 );
121
122 -- -----
123 -- COUNTRY + MOVIE_COUNTRY (M:N) - from production_countries API field
124 -- -----
125 CREATE TABLE IF NOT EXISTS country (
126     country_code    VARCHAR(10) PRIMARY KEY, -- ISO 3166-1 alpha-2 e.g. "US"
127     country_name    VARCHAR(255) NOT NULL
128 );
129
130 CREATE TABLE IF NOT EXISTS movie_country (
131     movie_id      BIGINT NOT NULL,
132     country_code   VARCHAR(10) NOT NULL,
133     PRIMARY KEY (movie_id, country_code),
134     FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE,
135     FOREIGN KEY (country_code) REFERENCES country(country_code) ON DELETE CASCADE
136 );
137
138 -- -----
139 -- REVIEW
140 -- -----
141 CREATE TABLE IF NOT EXISTS review (
142     review_id       VARCHAR(128) PRIMARY KEY,
143     movie_id        BIGINT      NOT NULL,
144     author_name     VARCHAR(255),
145     author_username  VARCHAR(255),
146     author_display_name VARCHAR(255),
147     author_avatar_path VARCHAR(255),
148     author_rating    DECIMAL(4, 1),

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149     content                TEXT                NOT NULL,
150     created_at              TIMESTAMP,
151     updated_at              TIMESTAMP,
152     review_url              TEXT,
153     FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE CASCADE
154 );
155
156 -- -----
157 -- TMDB_ACCOUNT (lightweight external user – no password column)
158 -- -----
159 CREATE TABLE IF NOT EXISTS tmdb_account (
160     account_id              BIGINT              PRIMARY KEY,
161     username                 VARCHAR(255),
162     display_name             VARCHAR(255),
163     include_adult            BOOLEAN,
164     avatar_path              VARCHAR(255)
165 );
166
167 -- -----
168 -- GUEST_SESSION
169 -- -----
170 CREATE TABLE IF NOT EXISTS guest_session (
171     guest_session_id         VARCHAR(255)        PRIMARY KEY,
172     expires_at               TIMESTAMP
173 );
174
175 -- -----
176 -- USER_RATING (from authenticated TMDB accounts)
177 -- -----
178 CREATE TABLE IF NOT EXISTS user_rating (
179     account_id              BIGINT              NOT NULL,
180     movie_id                BIGINT              NOT NULL,
181     rating_value             DECIMAL(3, 1)       NOT NULL,
182     created_at               TIMESTAMP,
183     PRIMARY KEY (account_id, movie_id),
184     FOREIGN KEY (account_id) REFERENCES tmdb_account(account_id) ON DELETE CASCADE,
185     FOREIGN KEY (movie_id)   REFERENCES movie(movie_id)           ON DELETE CASCADE
186 );
187
188 -- -----
189 -- GUEST_RATING (from guest sessions)
190 -- -----
191 CREATE TABLE IF NOT EXISTS guest_rating (
192     guest_session_id         VARCHAR(255)        NOT NULL,
193     movie_id                 BIGINT              NOT NULL,
194     rating_value             DECIMAL(3, 1)       NOT NULL,
195     created_at               TIMESTAMP,
196     PRIMARY KEY (guest_session_id, movie_id),
197     FOREIGN KEY (guest_session_id) REFERENCES guest_session(guest_session_id) ON DELETE
    CASCADE,

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198      FOREIGN KEY (movie_id) REFERENCES movie(movie_id) ON DELETE
      CASCADE
199 );
200
201 -- -----
202 -- Indexes for common query patterns
203 -- -----
204 CREATE INDEX IF NOT EXISTS idx_movie_release_date ON movie(release_date);
205 CREATE INDEX IF NOT EXISTS idx_movie_popularity ON movie(popularity DESC);
206 CREATE INDEX IF NOT EXISTS idx_movie_vote_average ON movie(vote_average DESC);
207 CREATE INDEX IF NOT EXISTS idx_review_movie_id ON review(movie_id);
208 CREATE INDEX IF NOT EXISTS idx_movie_cast_person_id ON movie_cast(person_id);
209 CREATE INDEX IF NOT EXISTS idx_movie_crew_person_id ON movie_crew(person_id);
210 CREATE INDEX IF NOT EXISTS idx_movie_keyword_kw_id ON movie_keyword(keyword_id);
211
```